Sheet 63 Thu. 29 Apr. 2027 Proof by contradiction and by case analysis

1. Warm-up · Flash [DIV] What is the parity of $n(n+1)$ for an integer n?

Ans. Always even: one of the two consecutive integers is even.

2. Core · Case analysis [DEM] Show that $n(n+1)$ is even for every integer n.

Ans. If n is even, the product contains an even factor. If n is odd, $n+1$ is even. In both cases, the product is even.

3. Core + · Contradiction [DEM] Show by contradiction that the sum of a rational r and an irrational number a is irrational.

Ans. Suppose $r+a=q\in\mathbb{Q}$; then $a=q-r\in\mathbb{Q}$ (difference of two rationals): contradiction with a irrational.

4. Back · Formula [IMG] $g(x)=2x+5$. Image of -1? Preimage of 11?

Ans. $g(-1)=3$; $2x+5=11\Leftrightarrow x=3$.

5. Running thread · Formula → graph [REG] Plot the points with x-coordinates -2; -1; 0; 1; 2 of the curve of $f(x)=x^2$; give their y-coordinates.

Ans. 4; 1; 0; 1; 4: the points are symmetric about the y-axis.

Sheet 64 Mon. 3 May 2027 Logical connectives, contrapositive and counterexample in context

1. Warm-up · Flash [DEM] Write the negation of "$x>2$", then of "$x>2$ or $x<-1$".

Ans. $x\leqslant 2$; $x\leqslant 2$ and $x\geqslant -1$, i.e. $-1\leqslant x\leqslant 2$.

2. Core · Method [DEM] Write the contrapositive of "if n^2 is even, then n is even".

Ans. "If n is odd, then n^2 is odd" (equivalent to the original statement).

3. Core + · Counterexample [DEM] Disprove: "every prime number is odd".

Ans. Counterexample: 2 is prime and even.

4. Back · True/false [PNP] True or false? Justify. The perimeter of a square is proportional to the length of its side.

Ans. True. $P=4c$: proportionality coefficient 4.

5. Running thread · Reading code [ALG] What does $h(3)$? Write $h(x)$ as an expression.

Ans. $h(3)=5\times 5=25$; $h(x)=(x+2)^2$.

Probability: conditioning and weighted trees

Sheet 65 Tue. 4 May 2027 Probability model and equally likely outcomes

1. Warm-up · Flash [PRB] A fair coin is tossed three times. How many outcomes?

Ans. $2^3=8$ equally likely outcomes.

2. Core · Method [PRB] Loaded die: $P(6)=0.25$ and the other five faces are equally likely. Calculate $P(1)$.

Ans. $\dfrac{1-0{,}25}{5}=0{,}15$.

3. Core + · Critical thinking [PRB] "It will either rain or not rain tomorrow: so there is a one-in-two chance." Is this equally likely model justified?

Ans. No: having two outcomes does not make them equally likely. Equally likely outcomes are an assumption of the model that must be justified.

4. Back · Flash [PER] Which shapes make up the net of a square-based pyramid?

Ans. 1 square and 4 triangles (isosceles).

5. Running thread · Translate [LIT] A square with side x has each of its sides increased by 3. Express the increase in area as a function of x.

Ans. $(x+3)^2-x^2=6x+9$.

Sheet 66 Mon. 10 May 2027 Law of large numbers (simplified version): simulation

1. Warm-up · Flash [STA] Out of 100 tosses, 47 "heads". What is the frequency of "heads"?

Ans. 0,47.

2. Core · Interpret [PRB] Why, over 10 tosses of a fair coin, can the frequency of "heads" move away from 0.5? What happens with 10,000 tosses?

Ans. Sampling fluctuation: small sample, high variability. Over large samples, the frequency stabilises near the probability 0.5 (law of large numbers).

3. Core + · True/false [PRB] True or false? Justify. Tossing a fair coin 1,000 times, we get exactly 500 "heads".

Ans. False. We obtain a number close to 500, rarely exactly 500.

4. Back · Problem [MOD] Price incl. VAT $P(x) = 1{,}2\,x$ (20% VAT). Calculate $P(250)$ and the preimage of 480.

Ans. $P(250) = 300$; $1{,}2x = 480 \Leftrightarrow x = 400$.

5. Running thread · Method [EQ] Solve in $\mathbb{R}$: $\dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

Ans. $x = 14$

Cross-multiplication: $2(2x-1) = 3(x+4)$.

Sheet 67 Tue. 11 May 2027 Python simulation of a repeated random experiment

1. Warm-up · Flash [PRB] $P(A) = 0.35$. Calculate $P(\overline{A})$.

Ans. $1 - 0{,}35 = 0{,}65$.

2. Core · Reading code [ALG] What does this program calculate? Approximately what value will it display?

```
from random import randint
compte = 0
for i in range(1000):
    if randint(1, 6) == 6:
        compte = compte + 1
print(compte / 1000)
```

Ans. The frequency of 6 in 1,000 rolls of a die; about $\dfrac{1}{6} \approx 0{,}167$.

3. Core + · Method [PRB] A fair 12-sided die numbered 1 to 12. Probability of obtaining a number prime?

Ans. Primes: 2; 3; 5; 7; 11: $\dfrac{5}{12}$ (1 is not prime).

4. Back · Construction steps [CON] M lies on the perpendicular bisector of $[AB]$. What can be said about MA and MB ? How can it be constructed with compasses?

Ans. $MA = MB$. Construction: two arcs of the same radius (greater than $\frac{1}{2}AB$) centred at A and B ; their intersections define the perpendicular bisector.

5. Running thread · Recognise [SOL] What kind of shape is the cross-section of a cube by a plane parallel to one of its faces?

Ans. A square with the same side as the face.

Sheet 68 Thu. 13 May 2027 Conditional probability $P_A(B)$

1. Warm-up · Flash [PRB] Of 25 half-boarders, 15 are girls. A half-boarder is chosen at random: probability that it is a girl?

Ans. $\dfrac{15}{25} = 0{,}6$.

2. Core · Method [PRB] In a class of 30 pupils, 12 play football, of whom 5 also play handball. A pupil is chosen at random: calculate $P_F(H)$, F : "plays football", H : "plays handball".

Ans. $P_F(H) = \dfrac{\text{Card}(F \cap H)}{\text{Card}(F)} = \dfrac{5}{12}$.

3. Core + · Method [PRB] In an equally likely model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate $P_A(B)$.

Ans. $\frac{20}{50} = 0{,}4$.

4. Back · Flash [NUM] Put in ascending order: $\frac{7}{4}, -\frac{3}{2}, 0{,}5$.

Ans. $-\frac{3}{2} < 0{,}5 < \frac{7}{4}$ $(-1{,}5 < 0{,}5 < 1{,}75)$.

5. Running thread · True/false [DEM] True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

Ans. False. Counterexample: a trapezium (not a parallelogram) has only two parallel sides.

Sheet 69 Tue. 18 May 2027 Conditional probability and two-way table

1. Warm-up · Flash [PRB] In the following table, what is the total of the "sport" row?

	Girls	Boys	Total
Sport	18	22	…
No sport	12	8	20
Total	30	30	60

Ans. $18 + 22 = 40$.

2. Core · Method [PRB] A pupil is chosen at random from this table. Calculate $P(S)$, $P_F(S)$ and $P_S(F)$ (F : girl, S : does sport).

Ans. $P(S) = \frac{40}{60} = \frac{2}{3}$; $P_F(S) = \frac{18}{30} = 0{,}6$; $P_S(F) = \frac{18}{40} = 0{,}45$.

3. Core + · True/false [PRB] True or false? Justify. In this table, $P_F(S) = P_S(F)$.

Ans. False. $0.6 \neq 0.45$: the reference sets (30 girls and 40 pupils who do sport) differ.

4. Back · Flash [PUI] Calculate $5^0 + 2^{-1}$.

Ans. $1 + \frac{1}{2} = \frac{3}{2}$.

5. Running thread · Method [REP] $ABCD$ is a parallelogram with $A(1;1)$, $B(5;2)$, $C(6;5)$. Determine D.

Ans. The diagonals have the same midpoint: $D = A + C - B = (2;4)$. Check: $\overrightarrow{AB} = \overrightarrow{DC} = (4;1)$.

Sheet 70 Thu. 20 May 2027 Weighted trees: construction

1. Warm-up · Flash [PRB] What is the sum of the probabilities on the branches leaving the same node?

Ans. 1.

2. Core · Method [PRB] In a population, 30% have a disease M. A test is positive (T) for 90% of those who are ill and for 5% of those who are not. Complete the branch probabilities of the tree.

Ans. $P(M) = 0{,}3$, $P(\overline{M}) = 0{,}7$; $P_M(T) = 0{,}9$, $P_M(\overline{T}) = 0{,}1$; $P_{\overline{M}}(T) = 0{,}05$, $P_{\overline{M}}(\overline{T}) = 0{,}95$.

3. Core + · Translate [PRB] Translate into symbols: "among those who are ill, 90 % test positive".

Ans. $P_M(T) = 0.9$: branch $M \to T$; do not write $P(T) = 0{,}9$.

4. Review · Flash [CAL] Calculate $\frac{2}{3} \div \frac{4}{9}$.

Ans. $\frac{2}{3} \times \frac{9}{4} = \frac{18}{12} = \frac{3}{2}$.

5. Running thread · Problem [DIV] Three lighthouses flash every 6, 8 and 9 seconds. They light up together at $t = 0$. When is the next simultaneous flash?

Ans. LCM $(6\,;\,8\,;\,9) = 72$: after 72 s.

Sheet 71 Mon. 24 May 2027 Weighted trees: multiplying along a path

1. Warm-up · Flash [CAL] Calculate $0.3 \times 0{,}9$.

Ans. $0{,}27$.

2. Core · Method [PRB] With the tree from sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.

Ans. $P(M \cap T) = 0{,}3 \times 0{,}9 = 0{,}27$; $P(\overline{M} \cap T) = 0{,}7 \times 0{,}05 = 0{,}035$.

We multiply along a path: P $(M \cap T) = P(M) \times P_M(T)$.

3. Core + · Method [PRB] Deduce $P(T)$.

Ans. Two paths lead to T : $0{,}27 + 0{,}035 = 0{,}305$.

4. Back · Factorise [DEV] Factorise $x^2 - 9$.

Ans. $(x-3)(x+3)$.

5. Running thread · Flash [GEO] The side $[BC]$ of length 13 cm is the hypotenuse of a right-angled triangle ABC. What is the radius of its circumcircle?

Ans. The midpoint of the hypotenuse is the centre: radius $= 13 \div 2 = 6.5$ cm.

Sheet 72 Tue. 25 May 2027 From the table to the tree and to symbolic notation

1. Warm-up · Flash [PRB] Express $\dfrac{30}{60}$ as a probability.

Ans. $0{,}5$.

2. Core · Method [PRB] From the table on sheet 69, build the tree (F or G, then S or $\overline{S}$) : give $P(F)$, $P_F(S)$ and $P_G(S)$.

Ans. $P(F) = 0{,}5$; $P_F(S) = \dfrac{18}{30} = 0{,}6$; $P_G(S) = \dfrac{22}{30} = \dfrac{11}{15}$.

3. Core + · Translate [PRB] Write symbolically and calculate "the probability of being a boy who does not do sport".

Ans. $P(G \cap \overline{S}) = \dfrac{8}{60} = \dfrac{2}{15}$; or $0.5 \times \dfrac{8}{30}$.

4. Back · Flash [QP] A car travels at constant speed: 150 km in 2 h. What distance in 5 h?

Ans. $75 \times 5 = 375$ km.

5. Running thread · Statement → formula [REG] A taxi charges a 2,000 Ar pick-up fee then 1,500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

Ans. $p(x) = 1\,500x + 2\,000$; $p(8) = 14{,}000$ Ar.

Sheet 73 Thu. 27 May 2027 Reversing the conditioning: P $_A(B)$ and $P_B(A)$

1. Warm-up · Flash [PRB] What is the difference between $P_M(T)$ and $P_T(M)$?

Ans. $P_M(T)$: probability of a positive test given that one is ill; $P_T(M)$: of being ill given that one is positive.

2. Core · Method [PRB] With the data from sheet 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.

Ans. $P_T(M) = \dfrac{P(M \cap T)}{P(T)} = \dfrac{0{,}27}{0{,}305} \approx 0{,}885$.

3. Core + · Method [PRB] Rare disease (1% of people ill); test positive for 99% of the ill and 5% of the not ill. Calculate $P_T(M)$.

Ans. $P(M \cap T) = 0{,}0099$; $P(\overline{M} \cap T) = 0{,}99 \times 0{,}05 = 0{,}0495$; $P(T) = 0{,}0594$; $P_T(M) = \dfrac{0{,}0099}{0{,}0594} = \dfrac{1}{6} \approx 0{,}17$.

Counter-intuitive result: a test that is "reliable to 99 % " is not enough.

4. Back · Graph [IMG] Curve of f :